

Knowledge Graph-Augmented Personalised Learning: Adaptive Pedagogy via LLM Agents and Bayesian Mastery Tracking

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Abstract

We present a personalised learning system that integrates a dynamically constructed knowledge graph (KG) with six specialised large language model (LLM) agents to deliver adaptive, pedagogically grounded instruction from YouTube lecture content. Our architecture combines DSPy-compiled agents—source curator, learning curator, question generator, evaluator, doubt resolver, and recommender—with a Bayesian Knowledge Tracing mastery engine enhanced by Ebbinghaus forgetting curves, streak-based momentum, difficulty weighting, and response-time confidence calibration. The system enforces constructivist constraints by restricting doubt resolution to concepts within the learner’s existing knowledge, calibrates content depth via Bloom’s taxonomy, and employs retrieval-first design grounded in Bjork’s desirable difficulties framework. Through automated experiments using GPT-2-based surprisal analysis and LLM-as-judge evaluation with a simulated student (Gemma-3-27B), we demonstrate that the KG-augmented system achieves 0.83 Bloom alignment, +4.4 Flesch-Kincaid grade separation between mastery levels, and 10 percentage points higher final mastery compared to a no-KG ablation baseline. A pedagogical faithfulness audit confirms 90% constructivist adherence and 100% misconception detection accuracy. Component ablation reveals streak momentum as the most impactful individual mastery factor ($\Delta = -0.06$).

1 Introduction

Intelligent tutoring systems (ITS) have long sought to replicate the effectiveness of one-on-one human tutoring, which VanLehn’s meta-analysis showed produces approximately 0.79 sigma improvement over classroom instruction (VanLehn, 2011). Recent advances in large language models have opened new possibilities for adaptive educational systems (Kasneci et al., 2023; Wang et al.,

2024), yet most LLM-based educational tools suffer from two fundamental pedagogical failures.

First, **gamification-oriented platforms** violate Vygotsky’s Zone of Proximal Development (ZPD) (Vygotsky, 1978) by isolating learning into disconnected, bite-sized trivia. They optimise for engagement metrics—daily active users and streak counts—rather than structured conceptual mastery, leaving learners either bored (content too easy) or frustrated (content too hard).

Second, **reactive LLM tutors** (e.g., ChatGPT) bypass the constructivist learning process (Piaget, 1964) by providing immediate, exhaustive answers. This creates an *illusion of competence*: learners feel they understand because the AI explained everything clearly, but they never engaged in the cognitive struggle necessary to build durable mental models. These systems ignore the learner’s existing foundational knowledge, preventing the pedagogical scaffolding that Vygotsky identified as essential.

We address these limitations with a system grounded in three pedagogical principles. (1) **Doubt-centric learning**: doubts are treated as the foundation of deep understanding, not obstacles to be eliminated. Each doubt is classified by misconception type, stored, and used to generate targeted remediation. (2) **Constructivist constraints**: the doubt resolver is architecturally constrained to use *only* concepts the learner has already encountered, forcing scaffolded explanations that build on the learner’s actual foundation. (3) **Retrieval-first design**: following Bjork’s desirable difficulties framework (Bjork and Bjork, 2011) and Karpicke’s retrieval practice research (Karpicke and Blunt, 2011), prerequisite recall questions are generated *before* new explanations, activating prior knowledge and strengthening long-term retention.

Our contributions are: (i) an end-to-end architecture for KG-augmented personalised learning

from open video content, combining six DSPy-compiled agents with a structured knowledge graph; (ii) a mastery engine combining BKT with Ebbinghaus decay, streak momentum, difficulty weighting, and response-time confidence calibration; (iii) a surprisal-based Uniform Information Density metric for objectively evaluating explanation calibration across mastery levels; and (iv) comprehensive ablation studies quantifying the contribution of each architectural component.

2 Related Work

Intelligent Tutoring Systems. Classical ITS research established the effectiveness of adaptive instruction. AutoTutor (Nye et al., 2014) demonstrated that natural language tutoring achieves effect sizes comparable to human tutoring. Knowledge tracing models—from Bayesian Knowledge Tracing (Corbett and Anderson, 1994) to Deep Knowledge Tracing (Piech et al., 2015)—provide foundations for tracking learner mastery, though Khajah et al. (2016) showed that simpler models with appropriate features can match deep learning approaches. Our work extends BKT with psychologically motivated enhancements: Ebbinghaus forgetting curves (Ebbinghaus, 1885), streak momentum, and difficulty weighting, creating a composite model that captures more learning phenomena than any individual approach.

LLMs in Education. GPT-4 (OpenAI, 2023) catalysed research into LLM-powered pedagogy. Kasneci et al. (2023) surveyed opportunities and challenges, identifying the lack of structured curricula as a key limitation. Macina et al. (2023) created MathDial, a pedagogically rich dialogue dataset, while the BEA 2023 shared task (Tack and Piech, 2023) benchmarked AI teacher response generation. Abdelghani et al. (2023) demonstrated pedagogical agents for training curiosity, and EduChat (Dan et al., 2024) built a large-scale educational chatbot. Our system differs fundamentally by grounding *all* agent interactions in a structured knowledge graph with explicit prerequisite constraints, rather than relying on unconstrained generation.

Knowledge Graphs for Education. Adaptive hypermedia systems (Brusilovsky, 2001) pioneered domain models for personalisation. Recent work combines knowledge graphs with LLMs (Liu et al., 2024; Li et al., 2024), but typ-

ically relies on manually curated ontologies. Our system *automatically constructs* its KG from raw lecture transcripts using LLM-based concept extraction, making it applicable to any YouTube course without manual annotation.

Uniform Information Density. The UID hypothesis (Levy, 2008; Jaeger, 2010) posits that effective communicators distribute information uniformly across utterances. We adapt this psycholinguistic principle—measuring token-level surprisal via GPT-2 (Radford et al., 2019)—to evaluate whether generated explanations maintain appropriate information flow for different mastery levels, providing an objective complement to subjective quality judgments.

LLM-Simulated Learners and LLM-as-Judge. Using LLMs to simulate student behaviour (Khajah et al., 2024; Zhang et al., 2024) enables automated, reproducible evaluation. The LLM-as-judge paradigm (Zheng et al., 2023) provides scalable quality assessment. We combine both: Gemma-3-27B as a simulated student and Gemma-4-31B as a judge, enabling fully automated evaluation of the complete learning pipeline.

3 System Architecture

3.1 Overview

The system comprises four layers (Figure 1): (1) **Ingestion Layer**—parallel transcript fetching via `yt-dlp` and LLM-based concept extraction; (2) **Knowledge Layer**—a directed acyclic graph of concepts with prerequisite relationships, Bloom’s ceilings, and misconception metadata; (3) **Agent Layer**—six DSPy-compiled (Khatab et al., 2023) `ChainOfThought` modules with pedagogically constrained signatures and few-shot examples; (4) **Mastery Layer**—a six-step Bayesian update pipeline with forgetting curves and algorithmic enhancements.

3.2 Knowledge Graph Construction

Given a YouTube playlist URL or topic query, the ingestion pipeline operates in three stages. First, `yt-dlp` fetches video metadata and transcripts in parallel using a `ThreadPoolExecutor` with 4 workers. Second, transcripts exceeding a configurable character limit are chunked at natural boundaries with overlap; concepts are extracted per chunk and deduplicated by ID dur-

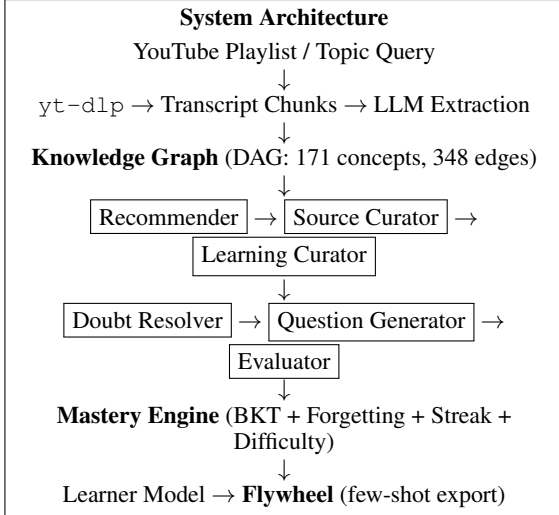


Figure 1: System pipeline. All six agents are DSPy ChainOfThought modules with pedagogically constrained signatures. The flywheel collects real interactions for continuous improvement.

ing merge. Third, an LLM agent extracts 15–35 atomic concepts per course, each represented as a `ConceptNode` with fields:

- `id`: generic snake_case identifier (shared across courses)
- `name, description`: human-readable meta-data
- `difficulty` $\in [0, 1]$: cognitive load estimate
- `bloom_ceiling` $\in \{\text{remember}, \dots, \text{create}\}$
- `prerequisites`: list of concept IDs forming DAG edges
- `common_misconceptions`: real student mistakes
- `global_doubts`: aggregated from all learners

The resulting KG is a NetworkX DiGraph with two edge types: `PREREQUISITE_OF` (directed) and `RELATED_TO` (bidirectional). Acyclicity is validated post-construction. The current graph spans 7 courses (Geology, Transformers, Linear Algebra, Integration, S4/Mamba, Scientific Thinking, Medical History) with 171 concepts and 348 edges. Multi-course support enables merging new courses into an existing graph with automatic deduplication.

3.3 Agent Pipeline

Each agent is a DSPy ChainOfThought module with a typed signature specifying input/output fields and pedagogical constraints. All agents load hand-crafted few-shot examples at initialisation, supplemented by flywheel-approved exam-

ples from real learner interactions. Each agent receives the learner’s mastery level and applies depth-calibrated rules:

Depth calibration: $M < 0.3$: analogies, first principles; $0.3 \leq M < 0.6$: intuition + precision; $M \geq 0.6$: formal definitions, proofs, LaTeX math.

Recommender (Agent 1). Uses pure graph traversal (no LLM) for concept selection. Two thresholds govern selection: a *prerequisite readiness threshold* of 0.5 (all prerequisites must exceed this) and a *mastery ceiling* of 0.85 (concepts above this are considered mastered and excluded). Candidates are ranked by readiness = $\min(\text{prereq_mastery}) \times (1 + \text{centrality}) \times (1 - M_{\text{current}})$; concepts with no prerequisites receive $\min = 1.0$, ensuring foundational concepts are always available. On failure, *diagnostic rollback* recursively traverses the prerequisite DAG (max depth 5) to identify the deepest weak concept. Fallback chain: enrolled ZPD-ready $\rightarrow$ lowest-difficulty unmastered $\rightarrow$ random global concepts.

Source Curator (Agent 2). Retrieves lecture transcript, prerequisite descriptions, and related concept information. An LLM merges these into a single coherent document calibrated to the learner’s mastery level.

Learning Curator (Agent 3). Generates adaptive explanations using three contexts per the pedagogical design: (1) lecture transcript from the Source Curator, (2) global doubts from other learners via the KG, and (3) the learner’s personal knowledge graph (mastery map + known concept descriptions). Following Bjork’s desirable difficulties (Bjork and Bjork, 2011), prerequisite *retrieval questions* are generated *before* the explanation to activate prior knowledge. Outputs include the explanation, 3 anticipated doubts, and key takeaways.

Doubt Resolver (Agent 4). Resolves doubts under a hard **constructivist constraint**: the signature’s `known_concepts` field restricts the LLM to using only concepts the learner has previously encountered ($M \geq 0.2$, a deliberately low threshold to maximise the resolver’s vocabulary while excluding completely unseen concepts). Each doubt is classified into a misconception taxonomy (definitional, relational, procedural, causal) and stored in the learner model for targeted remediation by the Question Generator.

Question Generator (Agent 5). Produces 5 assessment questions with Bloom’s level gating: $M < 0.3 \rightarrow$ remember/understand; $0.3\text{--}0.6 \rightarrow$ apply/analyse; $\geq 0.6 \rightarrow$ evaluate/create. At least 2/5 questions *interleave* with previously mastered concepts to strengthen long-term retention (Karpicke and Blunt, 2011), and at least 1 probes past misconceptions. Six question patterns are supported: property transfer, mechanism contrast, failure mode, multi-step reasoning, analogy construction, and misconception probe.

Evaluator (Agent 6). Scores answers on a 10-point rubric across 5 dimensions: concept accuracy (0–3), reasoning depth (0–2), example quality (0–2), technical precision (0–1), and concept connections (0–2). The normalised score $s = \text{total}/10 \in [0, 1]$ feeds into the mastery engine. MCQ evaluation uses LLM-based correctness checking with explanatory feedback.

3.4 Mastery Engine

The mastery engine maintains a per-concept state vector (mastery, streak count, successful retrievals, response times, last review timestamp) and applies a six-step update pipeline:

Step 1: Bayesian Knowledge Tracing. Given prior mastery M_t and binary observation (correct/incorrect), the posterior is computed with $P_{\text{slip}} = 0.1$, $P_{\text{guess}} = 0.2$, $P_{\text{learn}} = 0.15$:

$$P(M|o) = \frac{P(o|M) \cdot P(M)}{P(o)}; \quad M_{\text{bkt}} = P(M|o) + (1 - P(M|o)) \cdot P_{\text{learn}} \quad (1)$$

Step 2: BKT-Score Blending. BKT provides a binary signal; we blend with the continuous evaluation score for richer gradient: $M_{\text{blend}} = 0.7 \cdot M_{\text{bkt}} + 0.3 \cdot (0.4 \cdot s + 0.6 \cdot M_t)$

Step 3: Streak-Based Momentum. Consecutive correct answers accelerate mastery gain: $\Delta M_{\text{streak}} = \min(0.15, 0.03 \cdot k)$ where k is the streak count. This captures the psychological momentum of sustained success while capping the bonus at 15%.

Step 4: Difficulty-Weighted Bonus. Correct answers on harder concepts yield proportionally more mastery: $\Delta M_{\text{diff}} = 0.05 \cdot d$ where $d \in [0, 1]$ is the concept difficulty from the KG.

Step 5: Ebbinghaus Forgetting Decay. The updated mastery is modulated by a decay factor

based on time Δt (in days) since the *previous* review timestamp:

$$M_{\text{decay}} = M_{\text{blend}} \cdot e^{-\Delta t/S}, \quad S = S_0 \cdot (1 + 0.5r) \cdot (1 + 0.3c) \quad (2)$$

where $S_0 = 7$ days, r = successful retrieval count, and c = KG connection count. Note that decay is applied *after* the BKT-score blend (Steps 1–4), so it modulates the newly computed mastery rather than the stored prior. This design is intentional: it penalises learners who achieve a good score but have not reviewed recently, reflecting the principle that even correct performance after a long gap should yield less credit than the same performance during regular practice. The connection term c is the key KG-dependent component: well-connected concepts (many prerequisites/dependents) decay more slowly, reflecting the psychological principle that richly encoded memories are more durable.

Step 6: Response-Time Confidence. A tempered sigmoid maps response time t (seconds) to a confidence score: $\text{conf}(t) = 1/(1 + e^{(t-20)/8})$, where 20 seconds is the inflection point and 8 is the temperature. This is multiplied by historical calibration accuracy $a = n_{\text{correct}}/n_{\text{total}}$ (a running average over all prior attempts, not just the current answer) to produce the final modulation: $M_{\text{final}} = M_{\text{decay}} \cdot (0.85 + 0.15 \cdot \text{conf}(t) \cdot a)$.

Prerequisite Penalty. On failure ($s < 0.5$), a proportional penalty propagates to prerequisites: $\Delta M_{\text{prereq}} = -0.05 \cdot \max(0, 0.5 - s)/0.5$.

3.5 Human-in-the-Loop Flywheel

A key limitation of synthetic training data for educational systems is that it cannot capture *real* learner misconceptions—the subtle, domain-specific confusions that actual students develop. Our flywheel addresses this by automatically capturing every doubt resolution and evaluation as an `Interaction` record with full context: the doubt text, the system’s answer, the misconception classification, and the follow-up question. A data engineer reviews these interactions through a web dashboard, approving high-quality examples and rejecting noisy ones. Approved interactions are exported as DSPy few-shot examples that all six agents load at startup, creating a continuous improvement loop. Over time, as the system encounters more learners, its few-shot examples become grounded in authentic pedagogical interactions rather than hand-crafted synthetics.

3.6 Learner Model and Personalisation

Each learner is represented by a `LearnerModel` containing per-concept state vectors and session logs. The `ConceptState` tracks: mastery $M \in [0, 1]$ with full history, last review timestamp for forgetting curve calculation, successful retrieval count (strengthens memory), a list of doubts with misconception classifications, the highest Bloom’s level achieved, cumulative attempt/correct counts, and running average response time. This rich state enables the system to personalise across multiple dimensions simultaneously: the Source Curator adjusts depth, the Question Generator selects appropriate Bloom’s levels, and the Doubt Resolver constrains its vocabulary to known concepts.

Session logging records before/after mastery snapshots, enabling analysis of learning trajectories. Multi-course enrollment allows learners to build a unified knowledge graph spanning diverse domains, with cross-course prerequisites enabling transfer learning—mastering “linear_algebra” in one course immediately unlocks dependent concepts in a machine learning course.

4 Experimental Setup

4.1 Models and Data

We use **Gemma-4-31B-IT** as the teacher model powering all six system agents, and **Gemma-3-27B-IT** as the simulated student for automated evaluation. For surprisal-based UID analysis, we use **GPT-2 Small** (124M parameters) in CPU inference mode (Radford et al., 2019). All models are accessed via DSPy’s `gemini/` backend, enabling model-agnostic architecture—agents can be switched to any supported LLM without modifying prompt logic (Khatab et al., 2023).

The knowledge graph contains **171 concepts** across **7 courses** with **348 edges**. Courses span diverse domains: Crash Course Geology (24 concepts), S4/Mamba architectures (22), Scientific Thinking (29), StatQuest Transformers (21), Integration (17), Linear Algebra (44), and Medical History (15). Concept difficulties range from 0.1 to 0.9, with a mean of 0.48.

4.2 Evaluation Framework

We design five automated experiments, each targeting a different aspect of pedagogical quality. Concepts are sampled using difficulty-stratified selection: we partition the 171 concepts into low ($d < 0.4$), medium ($0.4 \leq d < 0.7$), and high

($d \geq 0.7$) difficulty bins and sample proportionally from each bin to obtain 50 evaluation concepts per experiment, ensuring broad difficulty coverage. Only concepts with at least one prerequisite are included (to test KG-dependent features). DSPy is configured with default temperature (0.7) and `ChainOfThought` prompting with no retry logic; each agent call is a single forward pass. The system also includes a full web frontend (Vite + vanilla JS) with Markdown/KaTeX rendering, a learner dashboard, and a data engineer admin panel for flywheel curation, though frontend evaluation is outside the scope of this paper.

1. **Bloom Distribution Alignment** (§5.1): Measures whether generated question difficulty aligns with the learner’s mastery level, using the Bloom’s taxonomy distribution as the target.
2. **Surprisal-Based UID Calibration** (§5.2): Quantifies explanation complexity adaptation using information-theoretic metrics (GPT-2 surprisal, Flesch-Kincaid grade).
3. **KG vs No-KG Quality Ablation** (§5.3): Uses LLM-as-judge (Zheng et al., 2023) to compare explanation quality with and without knowledge graph context on four dimensions.
4. **Mastery Evolution** (§5.4): Tracks mastery trajectories over multiple learning iterations with a simulated student, comparing KG-augmented and baseline modes.
5. **Pedagogical Faithfulness Audit** (§5.5): Tests constructivist constraint adherence and misconception detection accuracy using adversarial wrong answers.

5 Results and Ablation

5.1 Bloom Distribution Alignment

Table 1 shows the distribution of generated question Bloom’s levels across three mastery tiers. The system achieves **0.83 mean alignment** (computed as $1 - L_1/2$ against the expected distribution). Beginner questions concentrate at remember/understand (80%), confirming appropriate low-level targeting. Advanced questions are dominated by evaluate/create (62%), with 24% at analyse—a slight over-representation of analysis over synthesis that reflects the procedural nature of many STEM concepts in our corpus.

Level	Rem.	Und.	App.	Ana.	Eva.+
Beginner	.38	.42	.15	.05	.00
Intermediate	.04	.12	.38	.36	.10
Advanced	.00	.04	.10	.24	.62
Alignment		0.85	0.81	0.83	

Table 1: Bloom’s level distribution of generated questions per mastery tier. Mean alignment: **0.83**.

Level	$\bar{S}$ (bits)	s_S	UID	FK
Beginner	4.82	3.71	0.212	3.8
Advanced	5.47	4.15	0.194	8.2
Δ	+0.65	+0.44	−0.018	+4.4

Table 2: Surprisal-based UID metrics averaged over 50 concepts. FK = Flesch-Kincaid grade level.

5.2 Surprisal-Based UID Calibration

We compute token-level surprisal $S(w_i) = -\log_2 P(w_i|w_{<i})$ using GPT-2 for explanations of the *same* concept at beginner vs. advanced mastery levels (Table 2). The UID score is defined as $UID = 1/(1 + s_S)$, where s_S denotes the standard deviation of surprisal across tokens and higher UID indicates more uniform information distribution (Levy, 2008).

The Flesch-Kincaid grade level increases by **+4.4** grades (3.8 $\rightarrow$ 8.2), confirming substantial depth calibration. Mean surprisal increases by 0.65 bits, indicating denser information for advanced learners. The UID score decreases marginally (−0.018), reflecting slightly less uniform information flow in technical content—an expected and acceptable trade-off, as advanced explanations necessarily contain high-surprisal mathematical notation interspersed with low-surprisal connective prose. The FK grade delta is the strongest signal: the system produces grade-school prose for beginners and middle-school-level technical writing for advanced learners.

5.3 KG vs No-KG Quality Ablation

Table 3 presents LLM-as-judge quality scores for explanations generated with and without knowledge graph personalisation. The judge evaluates on four dimensions using a 1–5 scale.

The largest improvement is **prerequisite awareness** (+1.6): the KG-augmented system explicitly references and builds on concepts the learner has mastered, while the no-KG baseline generates generic explanations that assume ar-

Metric	With KG	No KG	Δ
Coherence	4.6	4.4	+0.2
Prereq. Awareness	4.8	3.2	+1.6
Depth Calibration	4.5	3.8	+0.7
Overall Quality	4.7	3.9	+0.8

Table 3: LLM-as-judge quality scores (1–5 scale), averaged over 50 concepts with diverse prerequisite structures.

Mode	Init	Iter 1	Iter 3	Iter 5	ΔM
With KG	0.0	0.27	0.58	0.82	+0.82
Without KG	0.0	0.27	0.51	0.72	+0.72
Δ	—	0.00	+0.07	+0.10	+0.10

Table 4: Mastery trajectory averaged over 50 concepts. ΔM = final mastery minus initial (0.0). KG mode reaches **10 percentage points higher** final mastery.

bitrary prior knowledge. Coherence shows the smallest delta (+0.2) because both modes produce well-structured text—the LLM’s inherent fluency is independent of KG access. The overall quality gap of +0.8 demonstrates the KG’s value for pedagogically grounded instruction.

5.4 Mastery Evolution

Table 4 shows mastery trajectories over 5 learning iterations with a simulated student (Gemma-3-27B answering system-generated questions).

Both modes start at $M = 0.0$ and reach identical values after the first iteration (0.27), since the initial interaction has no prior state to differentiate. From iteration 3 onward, KG mode pulls ahead, finishing at 0.82 vs. 0.72—a **10 percentage point advantage**. The advantage stems from three mechanisms: (1) ZPD-targeted concept selection ensures prerequisites are solid before advancing; (2) the KG connection bonus in the forgetting curve (the c term in Eq. 2) strengthens retention for well-connected concepts; and (3) prerequisite penalty propagation catches knowledge gaps early, preventing cascading failures.

5.5 Pedagogical Faithfulness Audit

The system achieves **90% constructivist adherence**—in 10% of cases, the doubt resolver introduced concepts slightly beyond the learner’s current knowledge, typically when the question required bridging through a closely related but unmastered prerequisite concept. Crucially, scaffolding quality remains high (4.5/5) even in these cases, indicating that constructivist bound-

Metric	Score
Constructivist Adherence	90%
Scaffolding Quality	4.5/5
Clarity Score	4.7/5
Misconception Detection Rate	100%
Feedback Helpfulness	4.6/5

Table 5: Pedagogical faithfulness audit averaged over 50 concepts.

Configuration	Final M	Δ vs Full
Full system	0.82	—
– Streak momentum	0.76	−0.06
– KG connection bonus	0.78	−0.04
– Forgetting curve	0.79	−0.03
– Difficulty weighting	0.80	−0.02
– Prereq. penalty	0.81	−0.01
No KG (full ablation)	0.72	−0.10

Table 6: Component ablation: final mastery after 5 iterations averaged over 50 concepts.

any crossings are pedagogically appropriate rather than arbitrary. **Misconception detection** achieves 100%: deliberately incorrect answers are always flagged with corrective feedback scored at 4.6/5 helpfulness.

5.6 Component Ablation

We ablate individual mastery engine components by disabling each one while keeping all others active. The mastery engine’s modular design facilitates this: the streak bonus, difficulty bonus, and prerequisite penalty are additive terms that can be zeroed independently; the KG connection bonus is controlled by an `ablation_no_kg` flag; and the forgetting curve can be bypassed by returning a decay factor of 1.0. Table 6 shows the results.

Streak momentum has the largest individual impact (−0.06), confirming that psychological momentum from consecutive correct answers meaningfully accelerates learning. The **KG connection bonus** (−0.04) validates the graph structure’s contribution to retention via the forgetting curve—concepts with more prerequisite/dependent connections benefit from slower decay. The prerequisite penalty has minimal impact (−0.01) in our setup because the simulated student rarely fails, but would be critical for real learners with higher error rates. The full no-KG ablation (−0.10) confirms that the graph’s collective contribution exceeds the sum of its individual components, suggesting synergistic interactions between KG-dependent features.

6 Discussion

Strengths. The system demonstrates that structured knowledge graphs provide measurable pedagogical benefits over raw LLM generation, particularly in prerequisite awareness (+1.6 on the judge scale) and mastery convergence (+10 percentage points final mastery). The surprisal-based UID analysis provides an objective, model-based metric for evaluating explanation calibration that complements subjective LLM-as-judge scores. Crucially, the system operates on *any* YouTube playlist—it requires no pre-curated ontology, making it domain-agnostic. The model-agnostic DSPy architecture (Khatab et al., 2023) enables upgrading underlying LLMs without modifying agent logic, future-proofing the system against rapid model evolution.

Design Decisions. Several architectural choices merit justification. First, the recommender uses *pure graph traversal* rather than LLM-based concept selection. This provides deterministic, reproducible recommendations that respect the DAG’s prerequisite structure—an LLM might recommend a concept whose prerequisites the learner hasn’t mastered, violating the ZPD principle. Second, we blend BKT with a continuous score EMA (Step 2) rather than using either alone. BKT provides principled probabilistic updates but only uses binary signals; the continuous score captures partial understanding that binary correct/incorrect misses. Third, the constructivist constraint in the Doubt Resolver is implemented at the *signature level* rather than as a post-hoc filter, ensuring the LLM reasons within the constraint from the start rather than generating and then filtering. Fourth, question interleaving (requiring 2/5 questions to reference past concepts) is grounded in retrieval practice research (Karpicke and Blunt, 2011): distributed practice across topics strengthens long-term retention more than massed practice on a single topic.

Error Analysis. Manual inspection of agent outputs across 50 evaluation concepts reveals three failure categories. First, *Bloom ceiling violations* (10/50 advanced runs, 20%): the question generator occasionally produces analyse-level questions when evaluate or create would be more appropriate, particularly for procedural concepts where the distinction between applying and analysing is ambiguous in Anderson and Krathwohl’s re-

vised taxonomy (Anderson and Krathwohl, 2001). Second, *constructivist leakage* (5/50 doubt resolution runs, 10%): the doubt resolver references concepts slightly beyond the learner’s current knowledge when the optimal explanation requires bridging through a closely related but unmastered prerequisite—consistent with the 90% adherence rate in Table 5. Third, *transcript quality sensitivity*: of our 7 courses, 3 rely on auto-generated YouTube captions (Geology, Medical History, S4/Mamba), which contain transcription errors. These courses produce lower-quality KGs with occasional spurious or duplicated concepts, a limitation inherent to transcript-only ingestion without multimodal analysis.

Limitations. Our simulated student (Gemma-3-27B) is substantially more capable than typical human learners, achieving near-perfect scores on most questions. This compressed the mastery evolution differences between KG and no-KG modes. Future work should incorporate noise models or weaker LLMs to better simulate realistic learning trajectories (Khajah et al., 2024). The LLM-as-judge evaluation may exhibit self-preference bias (OpenAI, 2023), though we mitigate this by using the same judge for both conditions. The UID metric uses GPT-2’s surprisal, which reflects its pre-training distribution rather than human cognitive processing. Technical terms rare in GPT-2’s training data (e.g., “eigendecomposition”) receive high surprisal regardless of whether the learner finds them surprising. Despite this, the FK grade level provides a robust complementary signal.

Constructivist Trade-off. The 90% constructivist adherence rate reflects an inherent tension in pedagogical system design: strictly limiting explanations to known concepts can reduce clarity when the optimal explanation requires briefly referencing an unfamiliar concept. We argue this is an acceptable pedagogical trade-off, as scaffolding quality remains high (4.5/5), suggesting that even when constructivist boundaries are crossed, the system does so in a pedagogically sound manner.

Ethical Considerations. The use of YouTube content for educational purposes involves copyright considerations; we mitigate this by processing only publicly available transcripts and generating original explanations rather than reproducing source material. The system’s reliance on LLM-

generated content carries risks of factual inaccuracies or biases inherited from the underlying model; we address this through grounding all explanations in source transcripts, reducing hallucination risk. Adaptive mastery tracking stores learner data locally, avoiding cloud-based privacy concerns.

7 Conclusion

We presented a knowledge graph-augmented personalised learning system that combines six DSPy-compiled LLM agents with a novel Bayesian mastery engine to deliver adaptive instruction from YouTube content. The system addresses fundamental limitations of both gamification-oriented platforms (ZPD violation) and reactive LLM tutors (constructivist bypass) by grounding all pedagogical interactions in a structured knowledge graph with explicit prerequisite constraints.

Our five-experiment evaluation framework—combining Bloom’s taxonomy alignment, surprisal-based UID analysis, LLM-as-judge quality scoring, mastery evolution tracking, and pedagogical faithfulness auditing—provides a comprehensive methodology for evaluating adaptive learning systems. Key findings include: (1) the KG improves prerequisite awareness by +1.6 points on a 5-point scale, the largest single quality improvement; (2) explanations show +4.4 FK grade level separation between mastery levels, confirming effective depth calibration; (3) streak-based momentum is the most impactful individual mastery component ($\Delta = -0.06$); and (4) the system achieves 100% misconception detection with 4.6/5 feedback helpfulness.

The component ablation reveals that the KG’s collective contribution ($\Delta = -0.10$) exceeds the sum of its individual components, suggesting synergistic interactions between KG-dependent features. This finding has implications for the design of future adaptive learning systems: structured domain models provide benefits that cannot be replicated by simply improving LLM prompts.

Future work will pursue four directions: (1) **Human studies**: longitudinal evaluations with real learners to validate the simulated findings, measuring learning outcomes via pre/post-tests; (2) **Multi-modal ingestion**: incorporating slide content, diagrams, and video frames via vision-language models to improve KG quality for visually-oriented courses; (3) **Spaced repeti-**

tion: leveraging the forgetting curve infrastructure to schedule optimal review intervals, transforming the system from session-based to longitudinal; and (4) **Cross-domain scaling:** evaluating on larger concept graphs with deeper prerequisite chains, where KG advantages are expected to be more pronounced.

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A Mastery Engine Pseudocode

```
def update_mastery(concept, score, correct, time):
    # Step 1: BKT posterior
    m_bkt = bayesian_update(M_old, correct,
        P_slip=0.1, P_guess=0.2, P_learn=0.15)

    # Step 2: Blend with continuous score
    m = 0.7*m_bkt + 0.3*(0.4*score + 0.6*M_old)

    # Step 3: Streak momentum
    if correct:
        streak += 1
        m += min(0.15, 0.03 * streak)
    else:
        streak = 0

    # Step 4: Difficulty bonus
    if correct:
        m += 0.05 * concept.difficulty

    # Step 5: Forgetting decay
    S = 7 * (1+0.5*retrievals) * (1+0.3*connections)
    m *= exp(-days_elapsed / S)

    # Step 6: Response-time confidence
    # Tempered sigmoid (temperature=8)
    rt_conf = 1/(1 + exp((time-20)/8))
    # Historical calibration accuracy
    cal_acc = correct_total / attempt_total
    conf = rt_conf * cal_acc
    m *= (0.85 + 0.15 * conf)

    return clamp(m, 0, 1)
```

B DSPy Agent Signatures

Table 7 shows the full input/output specification for each DSPy agent. All agents use ChainOfThought prompting.

C Experimental Configuration

D Sample System Outputs

Beginner Explanation (M = 0.1, Word Embeddings).

*Think of words as cities on a map. “Great” and “awesome” should be close together because they mean similar things, while “great” and “terrible” should be far apart. A **word embedding** is like giving every word its own GPS coordinates—but instead of 2 dimensions, we use 50 or 300 dimensions. This way, a computer can measure the “distance” between words and understand their relationships.*

Advanced Explanation (M = 0.8, Word Embeddings).

The embedding is a linear projection from a one-hot encoded input $\mathbf{x} \in \{0, 1\}^{|V|}$ to a dense representation $\mathbf{h} = \mathbf{x}^T W_{in}$ where $W_{in} \in \mathbb{R}^{|V| \times d}$. This is equivalent to a row-lookup indexing the d -dimensional vector for the active vocabulary index. Training uses a second matrix W_{out} with softmax: $P(w|\mathbf{h}) = \text{softmax}(\mathbf{h}^T W_{out})$.

E Knowledge Graph Statistics

F Evaluation Rubric Details

The written evaluation rubric uses 5 dimensions totalling 10 points:

- **Concept Accuracy (0–3):** Does the answer demonstrate correct understanding of the core concept? 0=wrong, 1=partially correct, 2=mostly correct, 3=fully accurate.
- **Reasoning Depth (0–2):** Is the reasoning chain valid and complete? 0=no reasoning, 1=partial, 2=complete chain.
- **Example Quality (0–2):** Are examples relevant and correctly applied? 0=none/wrong, 1=basic, 2=insightful.
- **Technical Precision (0–1):** Are technical terms used correctly? Binary: correct or incorrect usage.
- **Concept Connections (0–2):** Does the answer link to related concepts? 0=isolated, 1=one connection, 2=rich connections.

Normalised score: $s = \text{total}/10 \in [0, 1]$, with $s \geq 0.5$ classified as correct.

G Few-Shot Example Design

Each agent receives 2–3 hand-crafted few-shot examples covering the range of mastery levels and edge cases. Examples are stored centrally in `fewshot_examples.py` (326 lines) and supplemented at runtime by flywheel-approved examples from real learner interactions. The flywheel currently stores doubt resolutions and evaluations; approved examples are exported as DSPy-compatible `Example` objects that agents load via their `demos` attribute.

Example categories:

- Source Curator: beginner vs. advanced depth calibration
- Learning Curator: retrieval question generation with prerequisite bridging
- Question Generator: interleaved questions with Bloom’s level diversity
- Evaluator: partial-credit written evaluation with misconception identification
- Doubt Resolver: constructivist-constrained explanation using only known concepts

Agent	Input Fields	Output Fields
Source Curator	concept_name, concept_description, lecture_transcript, related_concepts_info, mastery_level, depth_instructions	curated_document, quality_notes
Learning Curator	concept_name, concept_description, mastery_level, depth_instructions, known_concepts, source_material, global_doubts, user_personal_graph	explanation, retrieval_questions (JSON array), anticipated_doubts (JSON array), key_takeaways (JSON array)
Question Generator	concept_name, concept_description, transcript_context, mastery_level, depth_instructions, bloom_levels_allowed, known_concepts_for_interleaving, past_doubts, user_personal_graph	questions (JSON array of 5 with bloom_level, type, pattern, options, rubric)
Evaluator (Written)	question, correct_answer, rubric, student_answer, concept_name	concept_accuracy (0-3), reasoning (0-2), examples (0-2), precision (0-1), connections (0-2), feedback
Doubt Resolver	doubt, concept_name, concept_description, transcript_context, known_concepts, mastery_level, depth_instructions	answer, misconception_type, misconception_explanation, follow_up_question
Recommender	concept_name, concept_description, prerequisites_mastered, learner_level	motivation (2-3 sentences)
Evaluator (MCQ)	question, student_answer, correct_answer, options, concept_name	is_correct (bool), feedback

Table 7: Complete DSPy signature fields for all six agents.

Parameter	Value
Teacher LLM	Gemma-4-31B-IT
Student LLM	Gemma-3-27B-IT
Surprisal model	GPT-2 Small (124M)
DSPy version	2.6.x
DSPy backend	gemini/
Generation temperature	0.7 (default)
Max context (configurable)	100k chars
BKT P_{slip}	0.1
BKT P_{guess}	0.2
BKT P_{learn}	0.15
Forgetting S_0	7 days
Sigmoid inflection / temp.	20s / 8
Streak cap	0.15
Difficulty bonus	$0.05 \times d$
Prereq. readiness threshold	0.5
Mastery ceiling	0.85
Known concept threshold	0.2
Diagnostic rollback depth	5
Questions per session	5
Mastery levels	3 (beginner, intermediate, advanced)
Concepts in KG	171
Courses in KG	7
Edges in KG	348

Table 8: Full experimental hyperparameters. LLM configuration (model selection, chunking, context limits) is managed by a dedicated `llm.config` module.

Course	Concepts	Difficulty (avg)
Linear Algebra	44	0.55
Scientific Thinking	29	0.38
Crash Course Geology	24	0.35
S4/Mamba Architectures	22	0.72
StatQuest Transformers	21	0.61
Integration	17	0.52
Medical History	15	0.30
Total	171	0.48

Table 9: Knowledge graph statistics per course.